SYNTHETIC OFFLINE-CRITIC EXPERIMENT | 4 NOVEMBER 2026

D-Flow and QGF: completed comparison

What is the task? These are artificial "choose a point" puzzles. Each action is a point (x,y). A hidden scoring function rewards some locations more than others. The method succeeds when its chosen point is close enough to the specified good target. There is no robot or maze in the counts in this report.

20 task configurations, 20 fresh test starts per condition, 8 D-Flow variants, and both lambda settings. Each fixed lambda condition gets its own kappa, eta and iteration selection on 40 separate validation starts. All reported counts below are held out.

This is a two-dimensional, one-step bandit diagnostic with an analytical reference flow and a fitted critic. It is not a reproduction of the QGF paper or a multi-step RL benchmark.

D-Flow variant	lambda = 0	lambda = 1
Regular / full gradient / full endpoint	201/400	209/400
Regular / identity gradient / full endpoint	201/400	201/400
Regular / full gradient / Euler endpoint	126/400	171/400
Regular / identity gradient / Euler endpoint	128/400	155/400
Stochastic / full gradient / full endpoint	105/400	172/400
Stochastic / identity gradient / full endpoint	164/400	191/400
Stochastic / full gradient / Euler endpoint	93/400	149/400
Stochastic / identity gradient / Euler endpoint	113/400	146/400

Reference	Held-out success	Mean reward
Behavior reference (BC)	56/400	0.256
QGF, fixed kappa = 0.02	60/400	0.264
QGF, fixed kappa = 0.04	62/400	0.274
QGF, independently tuned kappa	166/400	0.570

Observed pattern: lambda = 1 improved the pooled success of seven D-Flow variants and tied one. The strongest observed stochastic condition reached 191/400; regular full-gradient/full-endpoint reached 209/400. Fixed small-kappa QGF is much weaker here than tuned QGF, so the tuned reference is essential.

WHAT WAS ACTUALLY RUN

Tasks, data and tuning protocol

Think of a map with good and bad places to put a dot. Q predicts the score for each possible dot, using examples of dots and their observed scores. D-Flow and QGF use that prediction to move their proposed dot. The five families below change the good locations, the examples available, or the errors in Q.

Family: four variants each	Task and coverage design
Mode selection	Two reward modes; change the better side and reference-flow offset.
Narrow and sparse targets	Smaller reward peaks and tighter success regions, including shifted vertical targets.
Curved action distributions	Bending and wavy transports couple the two action coordinates.
Missing coverage	Remove a source tail or a ball near a useful target from the offline observations. The reference prior can still generate those actions.
Controlled hidden OOD error	Remove an action-space ball, then inject a smooth false critic peak strictly inside it. The injected error is exactly zero at every observed training action.

Offline data and critic. Each task uses 650-850 fixed action/reward observations with small Gaussian reward noise. A smooth RBF-plus-affine critic is fitted by ridge regression. Rewards are known mixtures of peaks with a weak quadratic tail penalty. Q equals expected immediate reward in this terminal bandit; no Bellman bootstrapping is used.

Reference flow. A known invertible two-dimensional triangular transport creates multimodal and curved action distributions. The full transport and its Jacobian are analytical. The continuous path is H(t,z) = [1-h(t)]z + h(t)F(z), with h(t) = (t+t^2)/2. This makes the source Euler endpoint an approximation rather than exactly F(z).

Setting	Values / definition
Kappa: Q correction strength	0, 0.02, 0.04, 0.06, 0.08, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1, 2, 3, 5
Eta: D-Flow optimizer step size	0.005, 0.01, 0.03, 0.1
D-Flow iterations	1, 2, 4, 8, 16, 32, 64, 128, 256
Lambda and stochastic noise	Lambda = 0 and 1; stochastic temperature T = 1. Regular variants omit the noise term.
QGF	32 integration steps; fixed kappa = 0.02 and 0.04; an additional reference tunes kappa on the full grid.

Selection: independently maximize validation success count, then mean validation reward; ties prefer fewer iterations and smaller coefficients. The 40 validation starts use true simulator reward. The 20 fresh test starts never select settings. All methods share the same initial seeds within a split.

LAMBDA CHOSEN ON VALIDATION, NEVER ON TEST RESULTS

Every task: successes out of 20

D = regular; S = stochastic. First letter: F = full gradient, I = identity gradient. Second letter: F = full endpoint, E = Euler endpoint. D-Flow cells show successes/20 [chosen lambda]. All selected kappa, eta and iterations are supplied in the CSV tables.

Task	BC	D FF	D IF	D FE	D IE	S FF	S IF	S FE	S IE	Q .02	Q .04	Q tuned
Right mode	5/20	12/20 [0]	12/20 [0]	12/20 [1]	12/20 [1]	12/20 [1]	11/20 [1]	14/20 [1]	13/20 [1]	5/20	5/20	11/20
Left mode	5/20	13/20 [0]	13/20 [0]	12/20 [1]	12/20 [1]	13/20 [1]	13/20 [1]	14/20 [1]	11/20 [1]	5/20	5/20	13/20
Offset right mode	4/20	9/20 [0]	9/20 [0]	8/20 [1]	8/20 [1]	12/20 [1]	12/20 [1]	10/20 [1]	13/20 [1]	4/20	4/20	8/20
Offset left mode	2/20	11/20 [0]	11/20 [0]	10/20 [1]	8/20 [1]	16/20 [1]	12/20 [1]	10/20 [1]	14/20 [1]	2/20	2/20	10/20
Narrow right	1/20	9/20 [1]	8/20 [0]	4/20 [1]	4/20 [1]	2/20 [1]	2/20 [1]	4/20 [1]	3/20 [1]	1/20	1/20	4/20
Narrow left	2/20	14/20 [1]	14/20 [1]	9/20 [1]	9/20 [1]	4/20 [1]	7/20 [1]	4/20 [1]	3/20 [1]	3/20	3/20	9/20
Sparse upper right	0/20	10/20 [1]	8/20 [1]	5/20 [1]	5/20 [1]	4/20 [1]	3/20 [0]	2/20 [1]	1/20 [1]	0/20	0/20	4/20
Sparse lower left	0/20	7/20 [1]	7/20 [1]	4/20 [1]	4/20 [0]	4/20 [1]	4/20 [1]	3/20 [1]	3/20 [1]	0/20	0/20	4/20
Rising curve	3/20	11/20 [1]	11/20 [0]	9/20 [1]	8/20 [1]	13/20 [1]	8/20 [1]	6/20 [1]	5/20 [1]	4/20	4/20	8/20
Falling curve	9/20	12/20 [1]	12/20 [0]	14/20 [1]	10/20 [1]	12/20 [1]	9/20 [1]	3/20 [1]	9/20 [1]	9/20	10/20	11/20
Wavy rising curve	2/20	12/20 [1]	9/20 [0]	5/20 [1]	5/20 [0]	12/20 [1]	9/20 [0]	10/20 [1]	8/20 [1]	2/20	2/20	5/20
Wavy falling curve	3/20	8/20 [0]	8/20 [0]	7/20 [1]	7/20 [1]	11/20 [1]	7/20 [1]	9/20 [1]	8/20 [1]	3/20	4/20	7/20
Unobserved right tail	0/20	8/20 [0]	8/20 [0]	9/20 [1]	6/20 [0]	7/20 [1]	10/20 [1]	7/20 [1]	5/20 [1]	1/20	1/20	4/20
Unobserved left tail	2/20	8/20 [0]	8/20 [0]	6/20 [0]	5/20 [0]	5/20 [1]	16/20 [1]	6/20 [1]	10/20 [1]	2/20	2/20	7/20
Missing upper target	0/20	11/20 [1]	11/20 [1]	9/20 [1]	9/20 [1]	5/20 [1]	3/20 [1]	2/20 [1]	0/20 [0]	0/20	0/20	8/20
Missing lower target	2/20	12/20 [1]	12/20 [1]	9/20 [1]	6/20 [1]	1/20 [1]	7/20 [1]	1/20 [0]	3/20 [1]	2/20	2/20	12/20
False upper peak	7/20	15/20 [0]	15/20 [0]	16/20 [1]	16/20 [1]	14/20 [1]	15/20 [1]	14/20 [1]	14/20 [1]	7/20	7/20	15/20
False lower peak	2/20	6/20 [1]	6/20 [1]	5/20 [1]	5/20 [1]	15/20 [1]	15/20 [1]	7/20 [1]	6/20 [1]	3/20	3/20	5/20
False right peak	4/20	10/20 [1]	10/20 [1]	8/20 [1]	8/20 [1]	6/20 [1]	14/20 [1]	9/20 [1]	10/20 [1]	4/20	4/20	9/20
False left peak	3/20	12/20 [1]	12/20 [0]	12/20 [1]	12/20 [1]	4/20 [1]	13/20 [1]	11/20 [1]	7/20 [1]	3/20	3/20	12/20

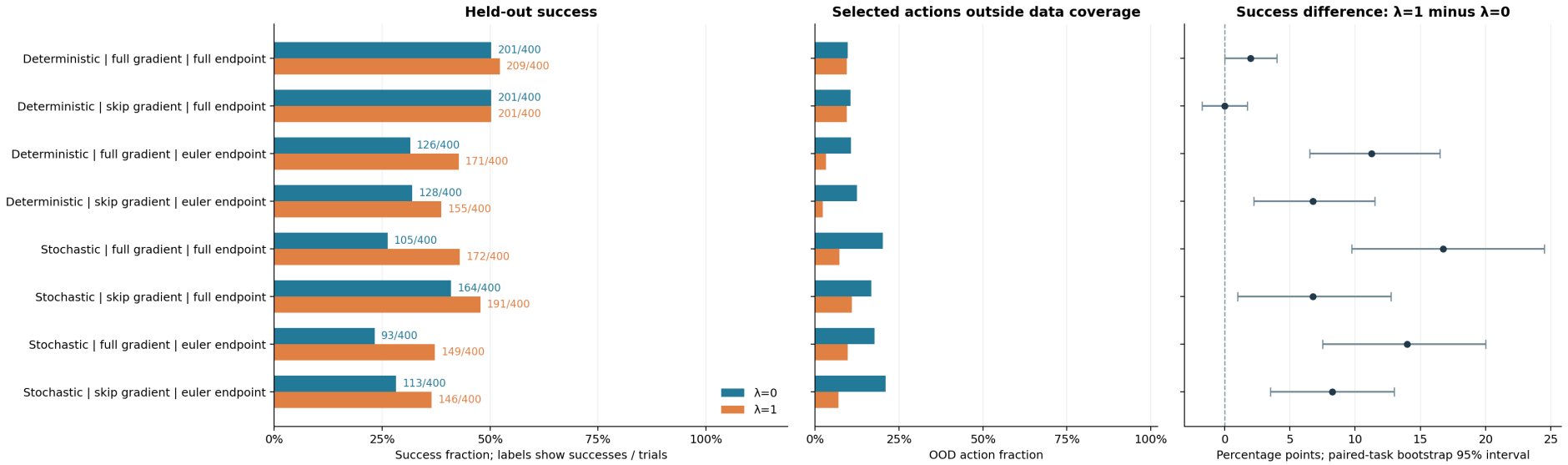
No tuned-QGF task has zero successes. Fixed small-kappa QGF gets 0/20 on three tasks, but tuning recovers 4/20, 4/20 and 8/20. These observations do not establish that QGF cannot solve the tasks.

OPTIONAL SHRINKING OF THE SOURCE NOISE INPUT

What lambda changes

D-Flow edits a random seed z and then generates the action F(z). Lambda multiplies an extra pull toward zero. At eta = 0.01, lambda = 1 replaces the old z contribution with 0.99z before adding the Q push and any noise. Lambda = 0 omits that shrinking. It does not measure or repair Q-function errors.

λ=1 adds a pull on the noise input toward zero



λ controls the optional pull toward the Gaussian source center; it does not correct Q-function errors. Each λ condition independently tunes κ, η and iterations on validation starts. This figure compares tuned conditions, rather than isolating a fixed-parameter causal effect of λ. Synthetic analytic-flow tasks; intervals exclude critic-training uncertainty.

Reading the comparison: the two lambda conditions independently retune kappa, eta and iterations. The figure therefore compares tuned conditions, rather than a fixed-parameter causal effect of lambda. OOD means low reference density or explicitly omitted coverage; a good missing-coverage target can itself be OOD.

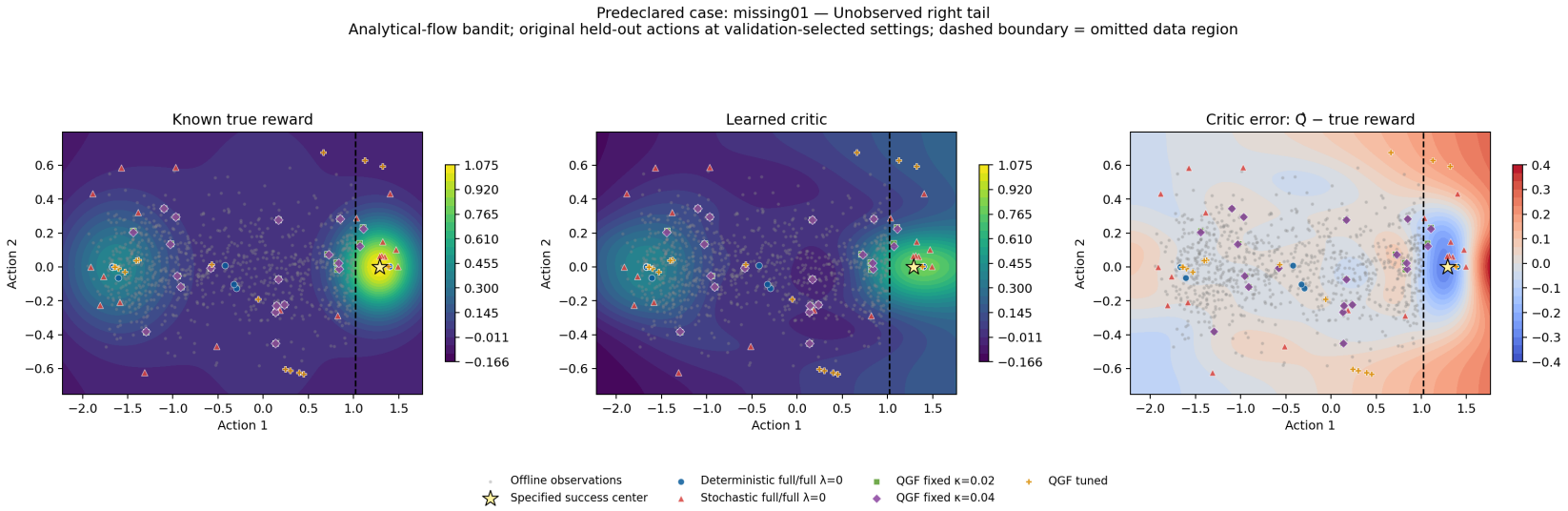
Exact update: $g(z) = J_E(z)^T$ grad Q(E(z)) for the full gradient, or grad Q(E(z)) for the identity gradient. Regular: z_next = z + eta [kappa g(z) - lambda z]. Stochastic: add sqrt(2 eta T) times a standard Gaussian draw. E is either F(z) or z+v(z,0). Final D-Flow actions always use F(z_final).

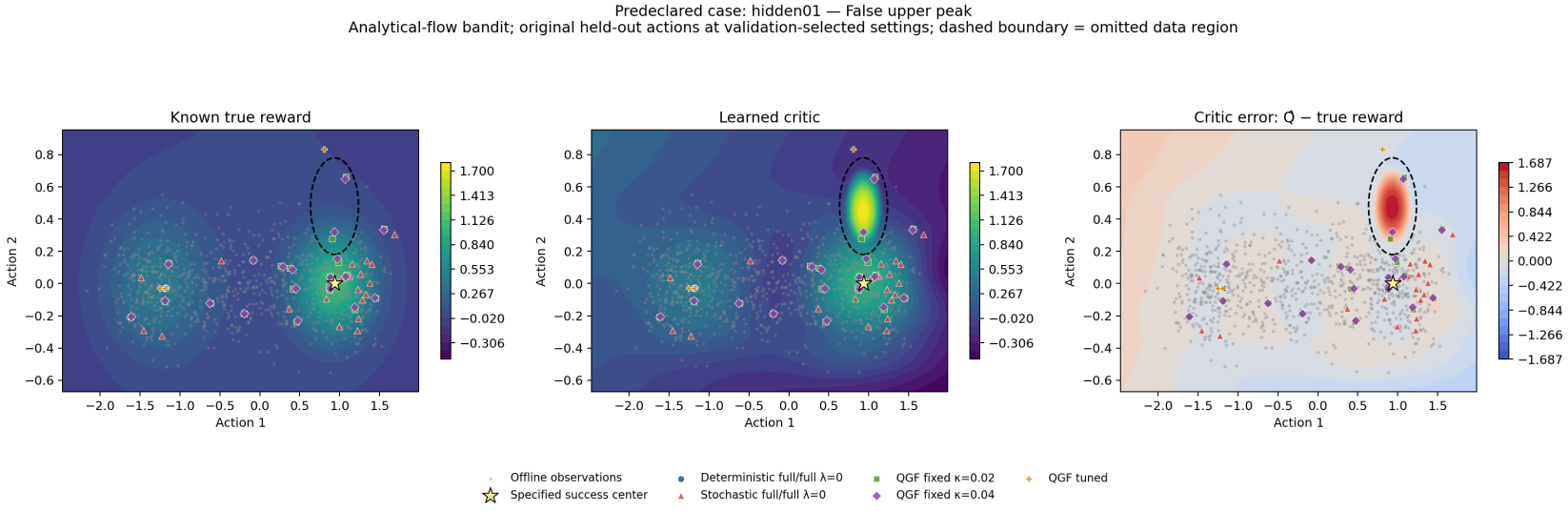
With lambda = 0, the stochastic method is finite-time noisy optimization. Only full-gradient/full-endpoint Langevin with the appropriate Gaussian prior term has the ideal BC-tilted sampling interpretation; finite step sizes and finite mixing still introduce error.

KNOWN REWARD, LEARNED CRITIC, AND CRITIC ERROR

Two predeclared unseen-action diagnostics

Dashed boundaries identify intentionally omitted data regions. Grey points are offline observations. Colored markers are original held-out actions at validation-selected settings. These cases were selected by task ID before examining outcomes.





WHAT THE EVIDENCE SUPPORTS

Verification, limits and reproduction

Separate next stage: a real OGBench robot task is being tested separately: move three cubes to their assigned goal positions. None of the point-selection success counts in this report represents a robot rollout. Robot results and the per-time-slice QGF code checks are saved in the separate official_pilot output folder.

Checks passed. All 173,160 validation rows and 8,000 test outcomes were checked. Hyperparameter and lambda choices match validation-only selection. All selected actions are finite. Every one of the 400 saved condition trajectories replayed exactly. Numerical flow and critic gradients were checked against finite differences; the largest critic-gradient check error was below 8e-8.

Additional controls. The output includes same-seed, fixed-parameter oracle-critic controls; removal of the injected hidden error; and independent ID/OOD critic-error probes. These controls do not retune hyperparameters or modify the main results.

Important limits. These are 20 inference starts per task and condition with one fitted critic per task, not 20 independent training seeds. True reward is available to the validation tuner, so the experiment does not solve fully offline hyperparameter selection. D-Flow receives a larger search space than QGF. The analytical full flow makes runtime comparisons unrepresentative of learned neural flows. Noise temperature is fixed rather than tuned.

Boundary selections. 219/320 D-Flow task/condition selections use kappa = 5 and 153/320 use 256 iterations. These are the best observed settings within the grid, not global optima. Wider sweeps may change the ranking.

Interpretation. On the controlled hidden-error family, stochastic identity-gradient/full-endpoint at lambda = 1 reaches 57/80, compared with tuned QGF at 41/80. Across the whole suite, regular full-gradient/full-endpoint remains stronger than the best stochastic condition. The results support a task-dependent search benefit, not correction of unknown critic values or general offline-RL superiority.

Reproduce the completed sweep

```
python source/run_comparison.py --output-dir rerun
python source/verify_comparison.py rerun
python source/comparison_diagnostics.py rerun
python source/plot_lambda_effect.py rerun
```

The bundle includes source code, dependencies, exact data/critic snapshots, task definitions, full validation grids, selected settings, individual held-out actions, verification records and figures. See README.txt and PROTOCOL_NOTES.txt for file descriptions.

Primary method references: Test-Time Gradient Guidance of Flow Policies in Reinforcement Learning (QGF); D-Flow: Differentiating through Flows for Controlled Generation. The eight combinations here are experimental variants, not eight algorithms claimed by those papers.